

The School of Mathematics



THE UNIVERSITY
of EDINBURGH

Working Title

by

Nolan Willoughby

Dissertation Presented for the Degree of
MSc in Operational Research

August 2026

Supervised by
Dr. Joerg Kalcsics

Abstract

this will come eventually

Acknowledgments

masive ups Joerg

Own Work Declaration

I attest that I am the sole author of this thesis and that all the content within it is my original work, except for any instances where I have clearly indicated otherwise in the text.

Contents

1	Introduction	1
2	Background	2
2.1	Home Healthcare	2
2.2	Application of OR to Healthcare Contexts	2
2.2.1	The Home Healthcare Routing and Scheduling Problem	2
2.2.2	Special Considerations	2
2.3	Relevant General Problem Structures	2
3	Problem Statement	3
4	Methodology	4
4.1	Design Considerations	4
4.2	Modelling Approach	4
4.3	Formulation	4
5	(Computational) Results	6
6	Conclusion	7
	Appendices	9
A	An Appendix	9
B	Another Appendix	10

List of Tables

List of Figures

1 Introduction

Section 2 will provide a review of relevant literature to the project in the interest of contextualizing the general nature of the work being done. Section 3 will formally define the problem being addressed by this work and the specific context of one home healthcare company. Section 4 will cover all approaches considered as potential means of addressing the problem, the specific approach chosen, and the way in which that approach was implemented, while Section 5 will provide detailed computational results of the methodology as implemented. Section 6 will conclude by briefly reviewing the accomplishments detailed throughout, detailing limitations of the work, and noting recommendations for future development of the work.

2 Background

2.1 Home Healthcare

Home Healthcare is the process by which certain health-related services are provided to patients in their home. These services typically involve nursing care, though can also consist of prescription and other medical supply delivery as well as assistance with day-to-day activity [1]. These services provided by home healthcare models are becoming increasingly essential as demographic changes yield older populations, greater geographical dispersion relative to both family and central hospital campuses, and a general preference toward aging in place. Additionally, home healthcare services can prove essential to modern healthcare systems by enabling personalized, expert care for patients with chronic illness, mobility challenges, and post-op recovery needs who may not otherwise be able to receive such quality care [2, 5].

Consequently, the importance of home healthcare companies, i.e. companies providing these services, is of increasing social interest. These groups must meet several operational challenges to meet the needs of their client base, and must do so with efficiency to ensure competitive advantage in a market system. Among the challenges of these groups is the need to find ways to efficiently provide diverse services requiring differing numbers and types of caregivers [3, 5]. Not only must these services be provided by varying numbers of caregivers with various qualification, they must also be delivered within strict time windows according to patient care needs [3]. while meeting regulatory compliance requirements. Given the operational complexity and social importance of these operations, the operations of home healthcare companies are a domain with high-potential for operations research applications.

2.2 Application of OR to Healthcare Contexts

2.2.1 The Home Healthcare Routing and Scheduling Problem

2.2.2 Special Considerations

Due to the significant human impact of healthcare contexts, certain considerations ought to be made to ensure positive, equitable outcomes when applying OR to these contexts.

2.3 Relevant General Problem Structures

As previously discussed, the problem at hand, the HHRSP, combines several elements of textbook OR problem structures. The first of these is the Assignment Problem.

The next of these is the Vehicle Routing Problem. In particular, the version of this problem with Time Windows is relevant due to the time-sensitive nature of certain home healthcare services such as insulin administration.

Another relevant problem structure, at least in certain contexts, is the Synchronization Problem.

3 Problem Statement

4 Methodology

4.1 Design Considerations

4.2 Modelling Approach

4.3 Formulation

Sets

D	set of days to schedule caregivers
C^d	set of caregivers available on day d
$C_D^d \subseteq C^d$	set of driving caregivers available on day d
L	set of localities served

Parameters

d_{ij}	distance or drive time between caregivers i and j
r_{jl}	distance or drive time between caregiver j and locality l
F_{ij}^d	number of pair visits on day d where both caregivers i and j are familiar to the client
f_{ij}^d	number of pair visits on day d where either caregiver i or j are familiar to the client
s_i^d	number of solo visits on day d where caregiver i is familiar to the client
V^d	total visit minutes on day d
V_{Pl}^d	visit minutes in locality l on day d requiring a caregiving pair
V_{sl}^d	visit minutes in locality l on day d requiring a single caregiver
l_i	binary indicator whether caregiver i is located in locality l
K	maximum distance carers will travel for pairing

Decision Variables

v_{il}^d	variable representing the linearization of the product of binary variables $z_{il}^d y_i^d$
w_{ijl}^d	variable representing the linearization of the product of binary variables $z_{jl}^d x_{ij}^d$

$$x_{ij}^d = \begin{cases} 1 & \text{if caregiver } i \in C_D^d \text{ paired with caregiver } j \in C^d \setminus \{i\} \text{ on day } d \\ 0 & \text{else} \end{cases}$$
$$y_j^d = \begin{cases} 1 & \text{if caregiver } j \in C^d \text{ is unpaired (i.e. does solo visits) on day } d \\ 0 & \text{else} \end{cases}$$
$$z_{il}^d = \begin{cases} 1 & \text{if caregiver } i \in C^d \text{ assigned to locality } l \in L \text{ on day } d \\ 0 & \text{else} \end{cases}$$

$$\max \quad \sum_{d \in D} \sum_{j \in C^d} \left(s_j y_j^d + \sum_{i \in C_D^d \setminus \{j\}} \left(F_{ij}^d x_{ij}^d + f_{ij}^d x_{ij}^d \right) \right) \quad (4.1)$$

$$\min \quad \sum_{d \in D} \sum_{j \in C^d} \sum_{i \in C_D^d \setminus \{j\}} d_{ij} x_{ij}^d + \sum_{l \in L} r_{jl} \left(x_{ij}^d + y_j^d \right) \quad (4.2)$$

$$\text{s.t.} \quad y_j^d + \sum_{i \in C_D^d} x_{ij}^d = 1 \quad \forall j \in C^d \setminus C_D^d, \forall d \quad (4.3)$$

$$y_j^d + \sum_{i \in C_D^d \setminus \{j\}} x_{ij}^d + \sum_{k \in C_D^d \setminus \{j\}} x_{jk}^d = 1 \quad \forall j \in C_D^d, \forall d \quad (4.4)$$

$$\sum_{i \in C_D^d} \sum_{j \in C^d \setminus \{i\}} l_j x_{ij}^d + w_{ijl}^d - \sum_{l' \in L \setminus \{l\}} w_{ijl'}^d \leq \lceil |C^d| \frac{V_{Pl}^d}{V^d} \rceil + \epsilon \quad \forall d \in D, \forall l \quad (4.5)$$

$$\sum_{i \in C_D^d} \sum_{j \in C^d \setminus \{i\}} l_j x_{ij}^d + w_{ijl}^d - \sum_{l' \in L \setminus \{l\}} w_{ijl'}^d \geq \lfloor \frac{|C^d|}{2} \frac{V_{Pl}^d}{V^d} \rfloor - \epsilon \quad \forall d \in D, \forall l \quad (4.6)$$

$$\sum_{j \in C^d \setminus C_D^d} \left(l_j y_j^d + \sum_{i \in C_D^d} \left(l_i y_i^d + v_{il}^d - \sum_{l' \in L \setminus \{l\}} v_{il'}^d \right) \right) \leq \lceil |C^d| \frac{V_{sl}^d}{V^d} \rceil + \epsilon \quad \forall d \in D, \forall l \quad (4.7)$$

$$\sum_{j \in C^d \setminus C_D^d} \left(l_j y_j^d + \sum_{i \in C_D^d} \left(l_i y_i^d + v_{il}^d - \sum_{l' \in L \setminus \{l\}} v_{il'}^d \right) \right) \geq \lfloor \frac{|C^d|}{2} \frac{V_{sl}^d}{V^d} \rfloor - \epsilon \quad \forall d \in D, \forall l \quad (4.8)$$

$$x_{ij}^d = 0 \quad \forall i \in C_D^d, \forall j \in \{j \in C^d \mid d_{ij} \geq 1\} \quad (4.9)$$

$$w_{ijl}^d \geq 0 \quad \forall i \in C_D^d, \forall j \in C^d \setminus \{i\}, \forall l \in L, \forall d \quad (4.10)$$

$$w_{ijl}^d \leq x_{ij}^d \quad \forall i \in C_D^d, \forall j \in C^d \setminus \{i\}, \forall l \in L, \forall d \quad (4.11)$$

$$w_{ijl}^d \leq z_{jl}^d \quad \forall i \in C_D^d, \forall j \in C^d \setminus \{i\}, \forall l \in L, \forall d \quad (4.12)$$

$$z_{jl}^d + x_{ij}^d - 1 \leq w_{ijl}^d \quad \forall i \in C_D^d, \forall j \in C^d \setminus \{i\}, \forall l \in L, \forall d \quad (4.13)$$

$$v_{il}^d \geq 0 \quad \forall i \in C^d, \forall l \in L, \forall d \quad (4.14)$$

$$v_{il}^d \leq y_i^d \quad \forall i \in C^d, \forall l \in L, \forall d \quad (4.15)$$

$$v_{il}^d \leq z_{il}^d \quad \forall i \in C^d, \forall l \in L, \forall d \quad (4.16)$$

$$z_{jl}^d + y_i^d - 1 \leq v_{il}^d \quad \forall i \in C^d, \forall l \in L, \forall d \quad (4.17)$$

$$x_{ij}^d \in \{0, 1\} \quad \forall i \in C_D^d, \forall j \in C^d \setminus \{i\}, \forall d \quad (4.18)$$

$$y_i^d \in \{0, 1\} \quad \forall i \in C^d, \forall d \quad (4.19)$$

$$z_{il}^d \in \{0, 1\} \quad \forall i \in C^d, \forall l \in L, \forall d \quad (4.20)$$

5 (Computational) Results

6 Conclusion

lol we've got a long way to go before i start doing this

References

- [1] A. Al-Hamad, Y. M. Yasin, K. Metersky, K. M. Kokorelias, L. Yasin, and F. Afzal. Home healthcare services and interventions for older adults: An umbrella review of systematic reviews and meta-analyses. *Journal of Ageing and Longevity*, 5(3), 2025.
- [2] S. Atta, V. Basto-Fernandes, and M. Emmerich. A concise review of the home health care routing and scheduling problem. *Operations Research Perspectives*, 15:100347, 2025.
- [3] S. V. Begur, D. M. Miller, and J. R. Weaver. An integrated spatial dss for scheduling and routing home-health-care nurses. *Interfaces*, 27(4):35–48, 1997.
- [4] S. Ceschia, F. Da Ros, L. D. Gaspero, S. Mancini, V. Maniezzo, R. Montemanni, R. M. Rosati, and A. Schaerf. A unified formulation for home healthcare routing and scheduling problems. *International Transactions in Operational Research*, pages 1–47, 2025.
- [5] D. S. Mankowska, F. Meisel, and C. Bierwirth. The home health care routing and scheduling problem with interdependent services. *Health Care Management Science*, 17(1):15–30, Mar 2014.
- [6] M. Masmoudi, J. Euch, and P. Siarry. Home healthcare routing and scheduling: operations research approaches and contemporary challenges. *Annals of Operations Research*, 343(2):701–751, Dec 2024.

Appendices

A An Appendix

B Another Appendix

Some other stuff.